

Day 6 — Saṁkrānti

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define *saṁkrānti* as the Sun’s transit from one *rāśi* to the next, name **Makara Saṁkrānti** and **Karka Saṁkrānti** as the two solstice-related *saṁkrāntis*, and locate all 12 *saṁkrāntis* on a calendar.

Materials

- Yesterday’s *rāśi*-wheel (work-in-progress)
- The marked calendar from Days 2–3
- A list of regional *Makara Saṁkrānti* names (Pongal, Lohri, Bihu, Maghi, Uttarayan)
- Activity 02 worksheet

Vocabulary introduced today

- *saṁkrānti* (IAST: *saṁ-krānti*; lit. “transit, crossing”) — the moment the Sun crosses from one *rāśi* into the next. There are ~12 *saṁkrāntis* per year, one for each *rāśi*-boundary.
- *Makara Saṁkrānti* (IAST: *makara saṁkrānti*) — the *saṁkrānti* around **14 January** when the Sun enters *Makara* (Capricorn). Traditionally marks the start of *uttarāyaṇa* in the *nirayana* reckoning.
- *Karka Saṁkrānti* (IAST: *karka saṁkrānti*) — the *saṁkrānti* around **16 July** when the Sun enters *Karka* (Cancer). Traditionally marks the start of *dakṣiṇāyana* in the *nirayana* reckoning.

Lesson flow

Warm-up (5 min)

- Prompt: “Have you ever celebrated **Pongal**, **Lohri**, **Bihu**, or **Uttarayan**?” Likely many hands. “All of these fall around the same date — January 14. Why?”
- Take 2 responses. Then: “They’re all names for the same astronomical event: **Makara Saṁkrānti**. Today we learn what that is, astronomically.”

Core (20 min)

1. **Definition.** A *saṁkrānti* (transit) is a **moment**, not a day — the exact instant when the Sun’s ecliptic longitude crosses a multiple of 30°, leaving one *rāśi* and entering the next. The *day* on which this happens is named after the *rāśi* the Sun is entering.
2. **The 12 *saṁkrāntis* in order:**

#	<i>Saṁkrānti</i>	Sun enters	Approximate date
1	<i>Meṣa Saṁkrānti</i> (also <i>Mahā Viṣuva</i> / Tamil	Meṣa (Aries)	14 April

#	<i>Samkrānti</i>	Sun enters	Approximate date
	<i>Puthandu</i>)		
2	<i>Vṛṣabha Samkrānti</i>	Vṛṣabha (Taurus)	14 May
3	<i>Mithuna Samkrānti</i>	Mithuna (Gemini)	14/15 June
4	Karka Samkrānti	Karka (Cancer)	16/17 July
5	<i>Simha Samkrānti</i>	Simha (Leo)	16/17 August
6	<i>Kanyā Samkrānti</i>	Kanyā (Virgo)	16/17 September
7	<i>Tulā Samkrānti</i>	Tulā (Libra)	17/18 October
8	<i>Vṛścika Samkrānti</i>	Vṛścika (Scorpio)	16/17 November
9	<i>Dhanus Samkrānti</i>	Dhanus (Sagittarius)	15/16 December
10	Makara Samkrānti	Makara (Capricorn)	14/15 January
11	<i>Kumbha Samkrānti</i>	Kumbha (Aquarius)	12/13 February
12	<i>Mīna Samkrānti</i>	Mīna (Pisces)	14/15 March

3. **Two *saṁkrāntis* are especially significant:**

- **Makara Saṁkrānti** (~14 January) — entry into *Makara*. In the *nirayana* system, this is the traditional start of *uttarāyana*. Across India:
 - **Pongal** in Tamil Nadu (4-day harvest festival)
 - **Lohri** in Punjab and Haryana (bonfire eve)
 - **Bihu** in Assam (Magh Bihu)
 - **Maghi** in Punjab (Sikh new year)
 - **Uttarayan** in Gujarat (kite festival)
 - **Khichdi** in Uttar Pradesh and Bihar
 - **Suggi** in Karnataka
- **Karka Saṁkrānti** (~16 July) — entry into *Karka*. In the *nirayana* system, this is the traditional start of *dakṣiṇāyana*. Celebrated as **Dakṣiṇāyana day** in parts of Maharashtra and as the **Rath Yatra** preparations in Odisha (loosely tied).

4. **Reference to the *Bṛhat Saṁhitā*** (Middle-band paraphrase):

Varāhamihira (6th c. CE), in his encyclopedic *Bṛhat Saṁhitā*, dedicates chapters to the *saṁkrāntis* — listing their dates, their seasonal markers, and (in his cultural context) their religious significance. The astronomical core — the Sun crossing a *rāśi*-boundary — is what we extract here. — paraphrased from *Bṛhat Saṁhitā*, chapter on solar transits (see [sources.md](#)).

Activity (15 min) — Activity 02 *Makara Saṁkrānti* explainer

See [activities/activity-02-makara-sankranti.md](#) for the full sheet. In summary:

- In small groups of 3–4, students prepare a 2-minute explanation of *Makara Saṁkrānti* answering:
 - What is the astronomical event?
 - What date does it fall on?

- What regional festivals celebrate it?
 - What does it traditionally mark?
- Each group presents to one neighbouring group (pair-share). Last 3 minutes: one group presents to the full class.

Wrap-up (5 min)

- Exit prompt (verbal): “*Today is (today’s date). Which saṁkrānti is closest? Past or upcoming?*”
- Preview Day 7: “*Tomorrow — the BIG question. Why is Makara Saṁkrānti on 14 January and not on 21 December (the actual winter solstice)? It’s a 24-day gap. There’s a reason.*”

Student-facing content

Saṁkrānti = the Sun crosses into a new rāśi. There are 12 per year.

The two most important:

- **Makara Saṁkrānti** (~14 January) — Sun enters *Makara* (Capricorn). Marks the start of *uttarāyaṇa* in the *nirayana* (sidereal) system. Celebrated as Pongal / Lohri / Bihu / Maghi / Uttarayan across India.
- **Karka Saṁkrānti** (~16 July) — Sun enters *Karka* (Cancer). Marks the start of *dakṣiṇāyana*.

Each *saṁkrānti* is a calendar moment, named after the *rāśi* the Sun is entering. Local festivals attach to many of them.

Homework

- Add the 12 *saṁkrānti* dates to your *rāśi*-wheel — write the approximate date at each *rāśi*-boundary.
- If your family celebrates one of the *Makara Saṁkrānti* festivals (Pongal, Lohri, Bihu, etc.), ask one elder how they celebrate, and write 3 sentences. If your family doesn’t celebrate any of them, write 3 sentences from a classmate’s account.

Differentiation

- **One tier up:** Each *saṁkrānti* has a religious / cultural significance attached in traditional almanacs. List the 12 and their associated traditional names from a *pañcāṅga* or Wikipedia. Note: this is cultural-historical knowledge, not astronomy — keep the two separate.
- **One tier down:** Memorise just two — *Makara* (January) and *Karka* (July). That’s enough for Day 7.

Teacher notes

- Resist the urge to “explain Day 7” early. Hold the puzzle: *Makara Saṁkrānti* on Jan 14 vs winter solstice on Dec 21 = 24-day gap. Let students sit with the gap; tomorrow we resolve it.

- Be ready for a religious-festival drift. Acknowledge briefly that these are real festivals; redirect to the astronomy.
- The Varāhamihira reference is paraphrased — don't pretend to quote.

Citation(s) used in this lesson

- **Varāhamihira, *Bṛhat Saṃhitā*** (6th c. CE) — paraphrased reference to the chapters on *saṃkrāntis*. See `sources.md`.